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Awaiting Receipt of Applications (ARA) Internal Org Web Service Technical User Guide

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1. General Information

1.1 Purpose

The purpose of this document is to provide a general description and usage guide for the ARA Internal Org Web Service.

1.2 General Description

ARA Internal Org web service is available for use by Internal Org users to Initiate/Submit ARA through a web service call.

1.3 Security

Institutes/Centers (Internal Orgs) need certificate-based authentication to access the ARA Internal Org web service (ARA Internal Org WS). For instructions on configuring the certificate, please refer to the Internal S2S Guide for eRA Web Services. As indicated in Section 3 of the Internal S2S Guide for eRA Web Services, the certificate needs to be registered after it has been procured. This step is conducted by the Internal Org Coordinator. The ARA Internal Org WS uses digitally signed certificates to authenticate a request. A service account ('NCI_SA', for example) must be created and mapped to a certificate by the Internal Org which wants to use the service. A new role "ARA_INITIATE_WEBSERVICE_ROLE" and corresponding privilege has been created for initiating/submitting ARA through ARA Internal Org WS. The below role must be added to the Internal Org's SA account in ERA_USER_ROLES_T) to call the ARA Internal Org WS.

User	Role Name	Role Description
Internal Org Service Account (e.g. NCI_SA)	ARA_INITIATE_WEBSERVICE_ROLE	Role to initiate or submit ARA through web service

1.4 Rest Service Locations

Following are the rest end point URLs for ARA Internal Org WS for each environment.

Environment	URL
TEST	https://web.test.era.nih.gov/ara/api/araservice
STAGE	https://web.stage.era.nih.gov/ara/api/araservice
PROD	https://web.aic.era.nih.gov/ara/api/araservice

2. ARA Internal Org Web Service

2.1 AraInternalOrgRequest

The table below provides the field information following successful authentication into ARA Internal Org WS.

#	Element Name	Description	Rule	Error when rule is not met	Error Code
1	requestingIc	The two character value for the IC which calls the service (e.g. 'CA' for NCI).	Required When requestingOrgId is not provided	- Requesting IC cannot be empty - IC is invalid	- ara.webservice.validate.noRequestIc - ara.webservice.validate.invalidIc
2	requestingOrgId	The Internal Org Id	Required When requestingIc is not provided	- Requesting Org cannot be empty - Org is invalid	- ara.webservice.validate.noRequestOrg - ara.webservice.validate.invalidOrg
3	requestorUserId	The requestor's IMPACII user ID (initiator's ID), used to identify the requestor for auditing purposes.	Required	Requester Id is invalid.	ara.webservice.validate.invalidRequesterId
4	piCommonsId	The PI's Commons user id	Required	- PI Commons Id cannot be empty. - Invalid PI Commons Id.	- ara.webservice.validate.emptyPiCommonsId - ara.webservice.validate.invalidPiCommonsId
5	orgName	Organization Name	Required	Organization Name cannot be empty	ara.webservice.validate.noOrgName
6	araTypeCode	The type of the ARA	Required	- ARA Type cannot be empty - ARA type is invalid for	- ara.webservice.validate.noAraType - ara.webservice.validate.invalidAraType

#	Element Name	Description	Rule	Error when rule is not met	Error Code
				this request	
7	projectTitle	Project Title of the ARA	Required	Project Title is missing	ara.webservice.validate.noProjectTitle
8	receiptDate	Anticipated Receipt Date	Required	Receipt Date is in invalid format	ara.webservice.validate.invalidReceiptDateFormat
			MM/DD/YYYY	Receipt Date is in invalid format	ara.webservice.validate.invalidReceiptDateFormat
			Anticipated Receipt Date cannot be further than 120 days from now.	ARA Receipt Date is 120 days in future	ara.webservice.validate.araInFuture

2.2 AraInternalOrgResponse

The Web Service will return results using the following structure:

2.2.1 Response Elements

There will be one status for the entire data load. If there are errors or warnings, they will be listed after the status.

#	Element Name	Description
1	successful	True: The ARA is successfully initiated. False: The ARA is not initiated successfully.
2	arald	The created arald if 'successful' is true.
3	errorMessageList	The list of error ErrorMessageList

2.2.2 ErrorMessageList

ErrorTypeCode	Description
ERROR	Error
INFO	Information message
WARNING	Warning

Following are the error codes that are sent in the response objects by the web service. The calling system can use the error sent by the web service or it can create its own message using the error code and parameters.

Error Code	Error Text	Error Type	Error Parameters List
<i>ara.webservice.validate.noProjectTitle</i>	Project Title is Missing	ERROR	
<i>ara.webservice.validate.invalidReceiptDateFormat</i>	Receipt Date is in invalid format	ERROR	

<i>ara.webservice.validate.araInFuture</i>	ARA Receipt Date is 120 days in future	ERROR	

2.3 HTTP Errors

- HttpStatus.FORBIDDEN
- HttpStatus.BAD_REQUEST
- HttpStatus.NOT_FOUND
- HttpStatus.INTERNAL_SERVER_ERROR

3. YAML Definition

YAML File `arainternalorgservices.yaml` is given below:

```
openapi: "3.0.0"
info:
  title: 'ARA Internal Org Rest Services'
  description: ARA API to Initiate Internal Org ARA Request and Approve Internal Org ARA Request.
  version: '0.0.1'
servers:
  - url: 'https://web.aic.era.nih.gov/ara/api'
    description: 'Production URL'
  - url: 'https://web.stage.era.nih.gov/ara/api'
    description: 'Stage URL'
  - url: 'https://web.test.era.nih.gov/ara/api'
    description: 'Test URL'
  - url: 'https://web.dev.era.nih.gov/ara/api'
    description: 'Dev URL'
tags:
  - name: ara-internal-org-services-controller
    description: 'ARA Internal Org Services Rest Controller'
paths:
  /araservice/initiateAraRequest:
    post:
      summary: Initiates Internal Org ARA request
      operationId: initiateAraRequest
      tags:
        - ARA Internal Org Service
      requestBody:
        description: Initiate Internal Org ARA request input. Required fields - Initiate ARA request details
        required: true
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/AraInternalOrgRequest"
      responses:
        "200":
          description: 'Initiated Internal Org ARA Response'
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/AraInternalOrgResponse"
        "400":
          description: "Bad Request"
        "403":
          description: "Not Authorized to Initiate a ARA Request"
        "404":
          description: "The specified resource was not found"
        "500":
          description: "Internal Server Error"
```

```
/araservice/approveAraRequest:
  post:
    summary: Approve ARA request
    operationId: approveAraRequest
    tags:
      - ARA Internal Org Service
    requestBody:
      description: Approve Internal Org ARA request input. Required fields -
        Approve Internal Org ARA request
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/AraInternalOrgApproveRequest"
    responses:
      "200":
        description: 'Approve Internal Org ARA Request Successfully'
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/AraInternalOrgResponse"
      "400":
        description: "Bad Request"
      "403":
        description: "Not Authorized to Approve a ARA Request"
      "404":
        description: "The specified resource was not found"
      "500":
        description: "Internal Server Error"

components:
  schemas:
    AraInternalOrgRequest:
      type: object
      description: Initiate Internal Org ARA Request Details.
      properties:
        requestingIc:
          type: string
        requestingOrgId:
          type: long
        requestorUserId:
          type: string
          nullable: false
        piCommonsId:
          type: string
          nullable: false
        orgName:
          type: string
          nullable: false
        araTypeCode:
          type: object
          description: ARA Type Code
          $ref: "#/components/schemas/AraType"
          nullable: false
```

```
projectTitle:
  type: string
  nullable: false
receiptDate:
  type: string
  nullable: false
foa:
  type: string
  maxLength: 10
councilDate:
  type: string
  maxLength: 7
programClassCode:
  type: string
  maxLength: 8
reasonForAra:
  type: string
  maxLength: 500
grantApplTypeCode:
  type: string
  maxLength: 1
grantActivityCode:
  type: string
  maxLength: 3
grantIc:
  type: string
  maxLength: 2
grantAdminOrgId:
  type: long
grantSerialNum:
  type: string
  maxLength: 6
grantSupportYear:
  type: string
  maxLength: 2
grantSuffixCode:
  type: string
  maxLength: 4
AraInternalOrgResponse:
  type: object
  description: Initiated Internal Org ARA Response Details.
  properties:
    araId:
      type: long
    successful:
      type: boolean
    errorMessageList:
      $ref: "#/components/schemas/ErrorMessageList"
AraInternalOrgApproveRequest:
  type: object
  description: Internal org Approve ARA request Details.
  required:
    - araInternalOrgRequest
  properties:
    araId:
```

```
    type: string
    nullable: true
    araInternalOrgRequest:
      $ref: "#/components/schemas/AraInternalOrgRequest"
  AraType:
    type: string
    enum:
      - 30
      - 31
      - 32
      - 33
      - 34
      - 35
      - 36
      - 37
      - 38
      - 39
      - 40
    description: >
      ARA Type Code Definition:
      * 30 - Accept application that requests $500,000 or more
      * 31 - Request for an assignment from IC
      * 32 - Request for an assignment from IRG
      * 33 - Request for an assignment from PI
      * 34 - Request for an assignment from Other
      * 35 - Request for a waiver
      * 36 - Replacement pages
      * 37 - Material to add with application
      * 38 - Request for No Further Processing
      * 39 - Take to...
      * 40 - Other
  ErrorMessageList:
    type: object
    description: API Error List.
    properties:
      errorMessageList:
        type: array
        items:
          $ref: "#/components/schemas/ErrorMessage"
  ErrorMessage:
    type: object
    description: API Error.
    properties:
      errorType:
        type: object
        description: Error Type
        $ref: "#/components/schemas/ErrorCode"
      errorText:
        type: string
        description: Error Text
      errorCode:
        type: string
        description: Error Code
      errorParametersList:
        type: array
```

```
    items:
      type: string
      description: Error Parameters List
  ErrorCode:
    type: string
    enum:
      - ERROR
      - WARNING
      - INFO
    description: >
      Error Type Code Definition:
        * ERROR - Error
        * WARNING - Warning
        * INFO - Information
```

4. Sample JSON Data

Initiate ARA Payload

```
{
  "requestingIc": "CA",
  "requestingOrgId": 12500008,
  "requestorUserId": "CHARLESV",
  "piCommonsId": "CKYUEN",
  "orgName": "UNIV OF NEBRASKA",
  "araTypeCode": "33",
  "projectTitle": "Test Project Title",
  "receiptDate": "11/20/2024",
  "foa": "PA-10-268",
  "councilDate": null,
  "programClassCode": null,
  "reasonForAra": null,
  "grantApplTypeCode": "1",
  "grantActivityCode": "R01",
  "grantIc": "MH",
  "grantAdminOrgId": 12500050,
  "grantSerialNum": "92436",
  "grantSupportYear": "1",
  "grantSuffixCode": "A1"
}
```

Approve ARA Payload

```
{
  "araId": "83047",
  "araInternalOrgRequest":
    {
      "requestingIc": "CA",
      "requestingOrgId": 12500008,
      "requestorUserId": "CHENSC",
      "piCommonsId": "CKYUEN",
      "orgName": "UNIV OF NEBRASKA",
      "araTypeCode": "33",
      "projectTitle": "Test Project Title",
      "receiptDate": "07/03/2024",
      "foa": "PA-10-268",
      "councilDate": null,
      "programClassCode": null,
      "reasonForAra": null,
      "grantApplTypeCode": "1",
      "grantActivityCode": "R01",
      "grantIc": "MH",
      "grantAdminOrgId": 12500050,
      "grantSerialNum": "92436",
      "grantSupportYear": "1",
      "grantSuffixCode": "A1"
    }
}
```

5. Workflow

There are two workflows to submit an ARA

- (1) Step1, Initiate an ARA by calling method `initiateAraInternalOrgRequest`
Step 2, Approve the ARA and updating the ARA by calling `approveAraInternalOrgRequest()`
Note: When sending the approve request, the information of the ARA still needs to be provided no matter there's any change or not.
- (2) The Referral Liason can approve an initiated ARA, or can submit an ARA directly by calling `approveAraInternalOrgRequest()` with empty `araId`.